

Exercise 8

Table: employees Advanced SQL Null Functions

employee_id	name	department	Salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

1. Show all employees with their Salary if Salary is NULL, display 0.

```
SELECT employee_id, name,
       IF NULL (Salary, 0) AS Salary_with_default
FROM employees;
```

employee_id	name	Salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Table: orders

order_id	customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

3. Find orders with NULL customer_id using IS NULL().

```
SELECT order_id, customer_id
FROM orders
WHERE ISNULL(customer_id);
```

Order-id	Customer-id
103	NULL

4. Show all orders. Replace NULL delivery_date with 'Pending'.

```
SELECT order_id, customer_id,
IFNULL(CAST(delivery_date AS CHAR), 'Pending')
AS delivery_status
FROM orders;
```

Order-id	Customer-id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 4: Products

Product-id	Name	Price	Discount
503	Mouse	12	NULL
502	Mouse	18	5
501	Keyboard	25	NULL

Table 3: Students

Student-id	Name	Grade
1	Ethan	85
2	Mayer	NULL
3	Olivia	90

5. Show all Students and their grades. Replace NULL with 0.

```
SELECT Student-id, Name,
       IF NULL (grade, 0) AS final-grade
FROM Students;
```

Student-id	Name	Final-grade
1	Ethan	85
2	Mayer	0
3	Olivia	90

6. Count students who haven't been graded

```
SELECT COUNT(*) AS not-graded-count
FROM Students
WHERE grade IS NULL;
```

Not-graded-count
1

Table 4: Products

Product - Id	Name	Price	discount
S01	keyboard	25	NULL
S02	mouse	15	5
S03	Monitor	100	NULL

7. Show Product-name, Price, and final Price after discount
(assume 0 if discount is NULL)

SELECT Product-id, Name,
Price - IFNULL (discount, 0) AS final-Price
FROM Products;

Product - Id	Name	Final-Price
S01	keyboard	25
S02	Mouse	10
S03	Monitor	100

Table: customers

customer_id	name	email
1	Linda	NULL
2	Joseph	Joseph@mail.com
3	Nia	NULL

8. Count how many customers have no email

```
SELECT COUNT(*) AS missing_email_count
FROM customers
WHERE email IS NULL;
```

missing_email_count
2

9. Show all customers with email. IF NULL, display "No Email"

```
SELECT customer_id, name,
       IF NULL (email, 'No Email') AS email-display
FROM customers;
```

customer_id	name	email-display
1	Linda	No email
2	Joseph	Joseph@mail.com
3	Nia	No email

Tables (customers)

Customerid	name	email
1	Linda	NULL
2	Joseph	Joseph@gmail.com
3	Joe	NULL

Table 6: Payments

Payment-id	method	Status
301	Credit	NULL
302	Paypal	Success
303	NULL	Failed

10. Show payment details with method replaced by "unknown" if NULL

```
SELECT Payment-id
       IFNULL (method, 'unknown') AS method-display,
       Status
FROM payments;
```

Payment-id	method-display	Status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

Table 8: Employees - Extra

emp-id	name	salary
1	NULL	100
2	NULL	200

Table 7: Inventory

Item-id	Item-name	Quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

11. Show items and their quantity (0 if NULL)

```
SELECT item-id, Item-name,
       IFNULL (Quantity, 0) AS Quantity_checked
FROM Inventory;
```

Item-id	Item-name	Quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees - Extra

emp-id	bonus	commission
1	NULL	300
2	100	NULL
3	NULL	NULL

12. Show employee ID and the first available value among bonus or commission.

SELECT emp-id
COALESCE(bonus, commission) AS first-available-reward
FROM employees-extra;

emp-id	First available reward
1	300
2	100
3	NULL

Table 9: Classes

Class id	Subject	Room
11	Math	NULL
12	English	Lab A
13	English	NULL

13. Count classes that don't have a room assigned.

SELECT COUNT(*) AS No-Room-Count
FROM Classes
WHERE room IS NULL;

No-Room-Count

2.

Table 11 Bank Accounts

Account-id	Account-type	Balance
A1	Savings	1000
A2	Current	5000
A3	NULL	0

Table 10 Attendance

Student-id	date	Status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

10. Show attendance records with status. Replace NULL with "Not marked"

```
SELECT Student-id, date,
        IFNULL(status, 'Not marked') AS attendance-Status
FROM Attendance;
```

Student-id	date	attendance-Status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

Student-id	date	Status
1	2023-06-01	NULL
2	2023-06-01	Present
3	2023-06-01	Absent

Table 10: Attendance

Table 11: Bank Accounts

account-id	account-type	Balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

15 Show account ID, account type (or 'unknown') and balance (or 0)

```
SELECT account-id
       IF NULL (account-type, 'unknown') AS type-display,
       IF NULL (balance, 0)              AS balance-checked
FROM bank-accounts;
```

account-id	type-display	balance-checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12: Projects

Project_id	title	Start_date	end-date
1	website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data migration	NULL	NULL

16. Show all projects with a start date. If start-date is NULL, display 'TBD'

```
SELECT project_id, title,
       IFNULL(CAST(Start_date AS CHAR), 'TBD') AS Start_display
FROM Projects;
```

Project_id	title	Start_display
1	website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13. Reviews

review-id	Product-id	Comment	rating
1	S01	Great Product	4
2	S02	NULL	NULL
3	S03	WORKS FINE	3

17. Display reviews showing comment (or No comment) and rating (or 0)

```

SELECT review-id, Product-id,
       IF NULL (comment, 'No Comment') AS comment_display
       IF NULL (rating, 0)              AS rating_display
FROM   reviews;

```

R review-id	Product-id	Comment-display	Rating-display
1	S01	Great Product	4
2	S02	No Comment	0
3	S03	WORKS FINE	3

Table 10: Suppliers

Supplier_id	Name	Phone	alt-Phone
1	Global Goods	NULL	123456789
2	Best Supplies	987654321	NULL
3	Value Source	NULL	NULL

10. Show the Supplier Contact number. use COALESCE (Phone, alt-Phone, 'No Contact').

```
SELECT Supplier_id, Name,
       COALESCE (Phone, alt-Phone, 'No Contact')
       AS Contact-Number
FROM Suppliers;
```

Supplier_id	Name	Contact-Number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	No Contact

Table 15: user_settings

user_id	theme	Language	time zone
1	Null	English	Null
2	Dark	Null	UTC + 1
3	Null	Null	Null

19. Show all users and their preferences. Replace all NULLS with defaults:

- Theme → "Light"
- Language → "English"
- Time zone → "UTC"

```
SELECT user_id,
       IF NULL (theme, 'Light') AS theme_set,
       IF NULL (Language, 'English') AS language_set,
       IF NULL (time zone, 'UTC') AS time zone_set
FROM user_settings;
```

user_id	theme_set	Language_set	time zone_set
1	Light	English	UTC
2	Dark	English	UTC + 1
3	Light	English	UTC

Table 10 Maintenance

record_id	machine_id	Issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

20. Show maintenance log with:

- Issue → default to "unknown issue"
- technician → default to "Not Assigned"

```
SELECT record_id, machine_id,
       IF NULL (issue, 'unknown issue') AS issue_log,
       IF NULL (technician, 'Not Assigned') AS technician_name
FROM maintenance;
```

record_id	machine_id	Issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	unknown issue	Not Assigned
3	M103	Jammed	Alex.

he name that